techniques developed for its proof, the various ways in which minors are handled or constructed. Most of these have not even been touched upon here, yet they seem set to influence the development of graph theory for many years to come.

Exercises

1.$^-$ Let $\leqslant$ be a quasi-ordering on a set X. Call two elements $x, y \in X$ *equivalent* if both $x \leqslant y$ and $y \leqslant x$. Show that this is indeed an equivalence relation on X, and that $\leqslant$ induces a partial ordering on the set of equivalence classes.

2. Let $(A, \leqslant)$ be a quasi-ordering. For subsets $X \subseteq A$ write

$$\mathrm{Forb}_{\leqslant}(X) := \{\, a \in A \mid a \not\geqslant x \text{ for all } x \in X \,\}.$$

Show that $\leqslant$ is a well-quasi-ordering on A if and only if every subset $B \subseteq A$ that is closed under $\leqslant$ (i.e. such that $x \leqslant y \in B \Rightarrow x \in B$) can be written as $B = \mathrm{Forb}_{\leqslant}(X)$ with finite X.

3. Prove Proposition 12.1.1 and Corollary 12.1.2 directly, without using Ramsey's theorem.

4.$^-$ Show that the relation $\leqslant$ between rooted trees defined in the text is indeed a quasi-ordering.

5. Show that the finite trees are not well-quasi-ordered by the subgraph relation.

6. The last step of the proof of Kruskal's theorem considers a 'topological' embedding of T_m in T_n that maps the root of T_m to the root of T_n. Suppose we assume inductively that the trees of A_m are embedded in the trees of A_n in the same way, with roots mapped to roots. We thus seem to obtain a proof that the finite rooted trees are well-quasi-ordered by the subgraph relation, even with roots mapped to roots. Where is the error?

7. Extend Kruskal's theorem to trees whose vertices are labelled from a well-quasi-ordered set. The tree embedding is defined as before but in addition respects the ordering of the labels.

8. Are the connected finite graphs well-quasi-ordered by contraction alone (i.e. by taking minors without deleting edges or vertices)?

9.$^+$ Relax the minor relation by not insisting that branch sets be connected. Show that the finite graphs are well-quasi-ordered by this relation.

10.$^+$ Show that the finite graphs are not well-quasi-ordered by the topological minor relation.

11.$^+$ Given $k \in \mathbb{N}$, is the class $\{\, G \mid G \not\supseteq P^k \,\}$ well-quasi-ordered by the subgraph relation?